Dasari Abhilash

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Education

Indian Institute of Technology — Delhi, India
PhD in Department of biochemical engineering and biotechnology — *Sep 2020 - Oct 2026*

University of Hyderabad — Hyderabad, India
Integrated Masters of Science, Systems Biology; GPA: 7.73/10.0 — *July 2015 - May 2020*

Skills

- **Programming Languages**: R, Python, BASH
- **Single-Cell Transcriptomics**: End-to-end sc/snRNA-seq processing (QC, normalization, clustering, cell-type annotation), trajectory inference, and ligand-receptor signaling using Seurat, Scanpy, SingleCellExperiment, Monocle, Slingshot, and CellChat
- **Spatial Transcriptomics**: High-resolution spatial profiling across sequencing and imaging platforms (10x Visium, Xenium, CosMx, MERFISH), including spot deconvolution, spatial domain clustering, and histological image integration using SPOTlight, BANKSY, and Squidpy.
- **Transcriptomics & Alternative Splicing**: Differential gene and transcript-level expression, novel isoform discovery, and alternative splicing quantification (exon skipping, intron retention, mutually exclusive exons) using STAR, Kallisto, Salmon, DEXSeq, rMATS, LeafCutter, Sleuth and FLAIR.
- **Genomics**: WGS, WES, and targeted panel pipelines covering alignment, germline/somatic SNP and indel calling, functional effect annotation, and GWAS association testing using BWA, Bowtie2, samtools, GATK, bcftools, FreeBayes, VEP, ANNOVAR, and PLINK.
- **Multi-Omics Analysis**: Multi-modal data integration combining transcriptomics, proteomics (mass spectrometry, differential protein abundance), and untargeted/targeted metabolomics with clinical metadata via matrix factorization, batch-correction, and co-expression network modeling for patient stratification and translational biomarker discovery.
- **HPC & Workflow Management**: Development of scalable, reproducible analytical workflows on Linux/UNIX High-Performance Computing (HPC) clusters utilizing Slurm, Nextflow, Snakemake, Conda, and Git.
- **Healthcare AI & Data Privacy**: Air-gapped, on-premise serving of open-source LLMs (Qwen 27B via llama.cpp) on local GPU/HPC clusters to strictly protect patient health information and clinical data confidentiality; development of secure agentic frameworks for automated biomedical workflows without external cloud exposure.
- **Experimental Lab Techniques**: Gel Electrophoresis, Western Blot, Bacterial culture handling, Microscopy, PCR
- **Languages Known**: English, Hindi, Telugu

Publications

First / Co-first Author

- Tanwar, P., Bhattacharya, D., **Dasari, A.**, Bijwe, M., Rana, R., Gupta, I. and Minocha, S., 2024. Dynamic gene expression pattern in zebrafish gallbladder: an experimental study. *International Journal of Surgery*, 110(12), pp.7585-7589. (DOI)
- Srivastava, P.P., **Dasari, A.**, Kumar, S., Gupta, I., Jain, D. and Minocha, S., 2025. Epigenetic co-regulator HCF-1 promotes lung cancer via O-GlcNAcylation-dependent pathways. *Molecular Therapy Oncology*, 33(4). (DOI)
- Budhraja, A., Basu, A., Gheware, A., **Abhilash, D.**, Rajagopala, S., Pakala, S., Sumit, M., Ray, A., Subramaniam, A., Mathur, P. and Nambirajan, A., 2022. Molecular signature of postmortem lung tissue from COVID-19 patients suggests distinct trajectories driving mortality. *Disease Models & Mechanisms*, 15(5), p.dmm049572. (DOI)
- Basu, A., Budhraja, A., **Abhilash, D.** and Gupta, I., 2021. Novel omics technology driving translational research in precision oncology. *Advances in Genetics*, 108, pp.81-145. (DOI)

Second Author

- Crepin, L., **Abhilash, D.**, Mougenot, N., Trenquier, E., Lanthiez, F., Abdou, I., Mihoc, M., de Raymon-Cahuzac, A., Ponnaiah, M., Karkeni, E., Chabbat-Courrede, C., Gautier, E.L., Gupta, I., Lebreton, G., Leprince, P., Boissonnas, A., Hatem, S.N. and Suffee, N., 2026. Epicardial fat drives macrophage response in atrial cardiomyopathy. *Circulation Research*. (DOI)
- Saxena, V., **Abhilash, D.**, Budhraja, A., Dey, S., Tiwari, P., Mohan, A., Gupta, I. and Kumar, S., 2026. RNA sequencing reveals differential expression of circular RNAs in human small cell lung cancer. *Scientific Reports*. (DOI)
- Dhania, N.K., Chauhan, V.K., **Abhilash, D.**, Thakur, V., Chaitanya, R.K., Dutta-Gupta, S. and Dutta-Gupta, A., 2021. Gut-specific arylphorin mediates midgut regenerative response against Cry-induced damage in Achaea janata. *Comparative Biochemistry and Physiology Part B: Biochemistry and Molecular Biology*, 255, p.110600. (DOI)

Fellowships and Academic Achievements

- **CSIR-UGC**: Secured 97th, 115 rank in Joint CSIR-UGC Junior Research Fellowship and Eligibility for Lectureship (NET) Exam in DEC 2018 and 2019 respectively
- **IIT-JAM**: Secured 454 and 687 ranks in IIT-JAM Biological Sciences and Biotechnology streams respectively
- **GATE**: Secured 292 rank in GATE-2020 Life Science stream

Research Experience

Indian Institute of Technology Delhi, India
PhD Thesis Project (Bioinformatics) *Sept 2020 - Oct 2026*

Working with Dr. Ishaan Gupta on Dissecting the role of alternative splicing in respiratory and hepatobiliary pathology.

- Developed automated, reproducible end-to-end pipelines (R, Python, BASH) on Slurm-managed HPC environments for deep transcriptomic profiling, alternative splicing quantification, and differential expression analysis across multi-cohort clinical datasets.
- Performed extensive multi-omics integration connecting transcriptomic signatures with clinical metadata, resulting in peer-reviewed translational publications.
- Engineered downstream visualization modules and statistical testing frameworks to interpret complex variant and isoform-level biological phenomena.
- Designed and managed scalable bioinformatics infrastructure, optimizing hardware procurement, budgeting, and high-performance computing (HPC) cluster nodes for multi-omics pipelines.
- Architected a privacy-preserving, on-premise LLM inference stack (llama.cpp, Qwen 27B & 125B) on local GPU hardware; automated bioinformatics scripting and code generation via multi-agent workflows while ensuring strict data privacy and isolation.
- Collaborated continuously with multi-disciplinary clinical, pathological teams at AIIMS to translate computational findings into pathophysiological insights.

IHU-ICAN, Sorbonne University Paris, France
Student Exchange program *Feb 2025 - July 2025*

Working with Dr. Maharaja Ponniah and Dr Nadine Suffee on discovering novel biomarkers of atrial cardiomyopathy with imagery and tissue biology.

- Conducted high-resolution single-cell (scRNA-seq) and spatial transcriptomics data processing, quality control, normalization, and cell-type annotation to characterize cellular microenvironments in human atrial cardiomyopathy.
- Integrated spatial transcriptomics data with tissue histology and high-resolution imaging to map spatial niches, disease-specific cell states, and intercellular signaling pathways (CellChat).

University of Hyderabad Hyderabad, India
M.Sc Thesis Project (Genomics) *May 2019 - May 2020*

Working with Dr. Vivek T to discover the genes underlying tolerance against cry toxin in Achaea Janata.

University of Hyderabad Hyderabad, India
Summer Research Intern (Computational Biology) *May 2018 - June 2018*

Worked with Dr. P Manimaran in understanding the applications of Chaos Game Representation(CGR) in Life Science related fields, writing a python script for CGR.

University of Hyderabad Hyderabad, India
Summer Research Intern *May 2016 - June 2016*

Worked with Prof. S.Rajagopal in understanding the interactions of various metabolites with Human Serum Albumin (HSA) domains I,II and III. Using fluorescence spectroscopy and Surface Plasmon Resonance

Leadership Experience

- **CAIC-PG Convenor**: Elected as a Co-curricular and Academic Interaction Council (CAIC) PhD Convenor 2024-25, IIT Delhi
- **RSF Executive committee member**: Nominated as Research Scholar Forum (RSF) student executive committee member 2023-24

Additional Experience, Workshops & Symposium

- Speaker at Spatial transcriptomics workshop conducted at ICGEB. Aug-2024
- Poster presentation at Indian cancer genome atlas (ICGA - 2024)
- Attended Human Cell Atlas Asia 2023 Meeting.
- Oral and Poster presentation at DBEB's annual in-house symposium BIOSPHERE in 2023 and 2024.
- Won first prize in Hackbiome, a bioscience hackathon organised by the Bioscience Society, NIT Rourkela, April 2021.
- Participated in 2nd Workshop of Advanced Bioinformatics: Network Medicine February-2021 organised by LAVIS - Institute of Neurobiology-Juriquilla, RiaBio.
- Participated in GIAN workshop on "Applied photosynthesis: putting nature into work" September-2019 at the University of Hyderabad.

References

- **Dr. Ishaan Gupta** | Associate Professor, DBEB, IIT Delhi (PhD Supervisor) | **Email:** ishaan@iitd.ac.in
- **Dr. Nadine Suffee** | Permanent research scientist, Inserm, Paris, France. | **Email:** nadine.suffee@yahoo.com